Electron-Ion Path Integral Monte Carlo

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I. INTRODUCTION

Hydrogen is the simplest element in the Universe. It has a nucleus made of only one proton, the ion, and an orbiting electron. An interesting question is whether the laws of quantum statistical physics are able to reproduce the formation of the Hydrogen atoms given a plasma made of an equal number of electrons and protons, a Two-Component-Plasma (TCP). This problem has been widely studied in literature starting from the pioneering exact analytical study of the non quantum TCP in one dimension of Edwards and Lenard [1, 2] to reach the realistic first principles ground state Monte Carlo calculations of Ref. [3] and non zero temperature Path Integral Monte Carlo (PIMC) calculations of Refs. [4, 5] or the coupled electron-ion Monte Carlo [6] where the ground state properties of the electrons component are used to perform a PIMC calculation on the protons component only.

Then we start from a complitely ionized TCP with a number of electrons $N_e = N/2$ equal to the number of protons $N_p = N/2$ in a volume Ω and in thermal equilibrium at an inverse temperature $\beta = 1/k_B T$ with k_B Boltzmann constant and T the absolute temperature. We use a PIMC calculation [7] with a number $M = \beta/\tau$ of timeslices of length τ which discretize the imaginary time interval $[0, \beta[$ to determine the thermodynamic and structural properties of the plasma. In particular the structural clustering that we observe proves the ability of the quantum statistical physics theory to predict the atom formation from first principles when one introduces just Coulomb pair interactions. [1]

Since the proton is about 2000 times more heavy than the electron we expect the protons species paths to be much more contracted than the ones of the electrons species. But as we will see this occurrence does not pose any serious limitation to what can be simulated but only a slowing down of the computer time needed to measure the properties of the protons component.

We know that both the electron and the proton are spin 1/2 identical particles and as such they each obey to the Fermi-Dirac statistics. But as we will see there is just a slight difference between treating both species as fermions or as bosons. This is essentially due to the property of bosons to 'like' themselves and of the fermions to 'dislike' themselves. Of course while bosons simulations are exact, fermions simulations are affected by the so called 'fermions sign problem' [8] which can be overcome only by approximate methods like the Restricted PIMC (RPIMC) [9].

Due to the divergence of the Coulomb potential at the origin like $1/r$ it is necessary to introduce a hard core σ for the nuclei, the protons, while we will consider the electrons as dimensionless and pointwise. This is needed in order to obtain a stable many body system that does not collapse. As we will see, for a given temperature T, the minimum hard core diameter preventing the collapse will depend on temperature, $\sigma = \sigma(T)$.

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[1] It would be interesting to see if, for more complicated atoms, the introduction of nuclear forces would still make it possible to predict the atom formation.

Let us consider a mixture of electrons, e, and protons, p. In reduced units we will consider $\hbar = k_B = 1$. We measure masses in units of the electron mass m_e so that the reduced mass of the electron will be $\mu_e = m_e/m_e = 1$ and the reduced mass of the proton $\mu_p = m_p/m_e \approx 1836.15$. The characteristic kinetic energy is

$$\mathscr{K} = \frac{\hbar^2}{2m_e\ell^2}, \tag{1.1}$$

with ℓ a length. The characteristic Coulomb energy is

$$\mathscr{V} = \frac{e^2}{\ell} = \frac{2\mathscr{K}\ell}{\mathscr{L}}, \tag{1.2}$$

with e the electron charge magnitude and

$$\mathscr{L} \approx 5.29177 \times 10^{-9}\text{cm} \tag{1.3}$$

a characteristic length. Then, if we denote with $n = N/\Omega$ the reduced number density this will correspond to a real number density of $n\mathscr{L}^{-3}$. For example for metallic hydrogen we have a mass density $\approx 1\text{g/cm}^3$ which corresponds to a reduced density of $n \approx \mathscr{L}^3/m_p \approx 0.0885943$.

The size of the nucleus of the hydrogen atom is $\approx 2 \times 10^{-13}$cm which in reduced units is $\sigma \approx 3.77945 \times 10^{-5}$. On the other hand we will consider the electrons as pointwise.

We may then take as characteristic temperature

$$\mathscr{T} = \frac{\mathscr{K}\ell^2}{\mathscr{L}^2 k_B} \approx 315775\text{K}. \tag{1.4}$$

So that for metallic hydrogen in planetary interiors an absolute temperature of $\approx 10^5$K will correspond to a reduced temperature of $T \approx 10^5/\mathscr{T} \approx 0.0316681$.

The Hamiltonian and the density matrix in reduced units will then read

$$H = -\sum_{\alpha=1}^{N} \frac{1}{2\mu_{i_\alpha}}\nabla_\alpha^2 + \sum_{\alpha<\beta=1}^{N} \phi_{i_\alpha i_\beta}(r_{\alpha\beta}), \tag{1.5}$$

$$\rho = e^{-H/T}. \tag{1.6}$$

where we indicate with a Greek index the particle label and with a Roman index the species label so that for example i_α is the species label of particle α, either 1 for the electrons or 2 for the protons, $\boldsymbol{r}_\alpha$ is the position vector of particle α, and $r_{\alpha\beta} = |\boldsymbol{r}_{\alpha\beta}| = |\boldsymbol{r}_\alpha - \boldsymbol{r}_\beta|$ is the distance between particles α and β. Distances are measured in units of $\mathscr{L}$. The reduced masses are then $\mu_1 = \mu_e$ and $\mu_2 = \mu_p$. The partial pair potential ϕ_{ij} is the usual Coulomb potential given by

$$\phi_{ij}(r) = \begin{cases} \epsilon_{ij}/r & r > \Sigma_{ij} \\ \infty & \text{else} \end{cases}, \tag{1.7}$$

where in the Two-Component-Plasma (TCP) we choose

$$\boldsymbol{\epsilon} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \tag{1.8}$$

$$\boldsymbol{\Sigma} = \begin{pmatrix} 0 & \sigma/2 \\ \sigma/2 & \sigma \end{pmatrix}. \tag{1.9}$$

Note that at a given T if we choose σ too small the whole system collapses with all the negative light particles falling on the heavy positive particles.

We will indicate with N_i the number of particles of species $i = 1, 2$ and $N = \sum_i N_i$ the total number of particles with $x_i = N_i/N$ the molar fraction for species i such that $\sum_i x_i = 1$.

From Fig. 1 we see the formation of clusters made of an heavy positive particle (of species 2) surrounded by a cloud of light negative particles (of species 1). We also see how changing the statistics between particles of the same species from Bose-Einstein to Fermi-Dirac makes them dislike themselves more as expected from the Pauli exclusion principle. This is confirmed by the snapshots of Fig. 2. From the snapshots we see how the electrons are in the quantum regime with extended paths and the protons are in the classical regime with contracted paths.

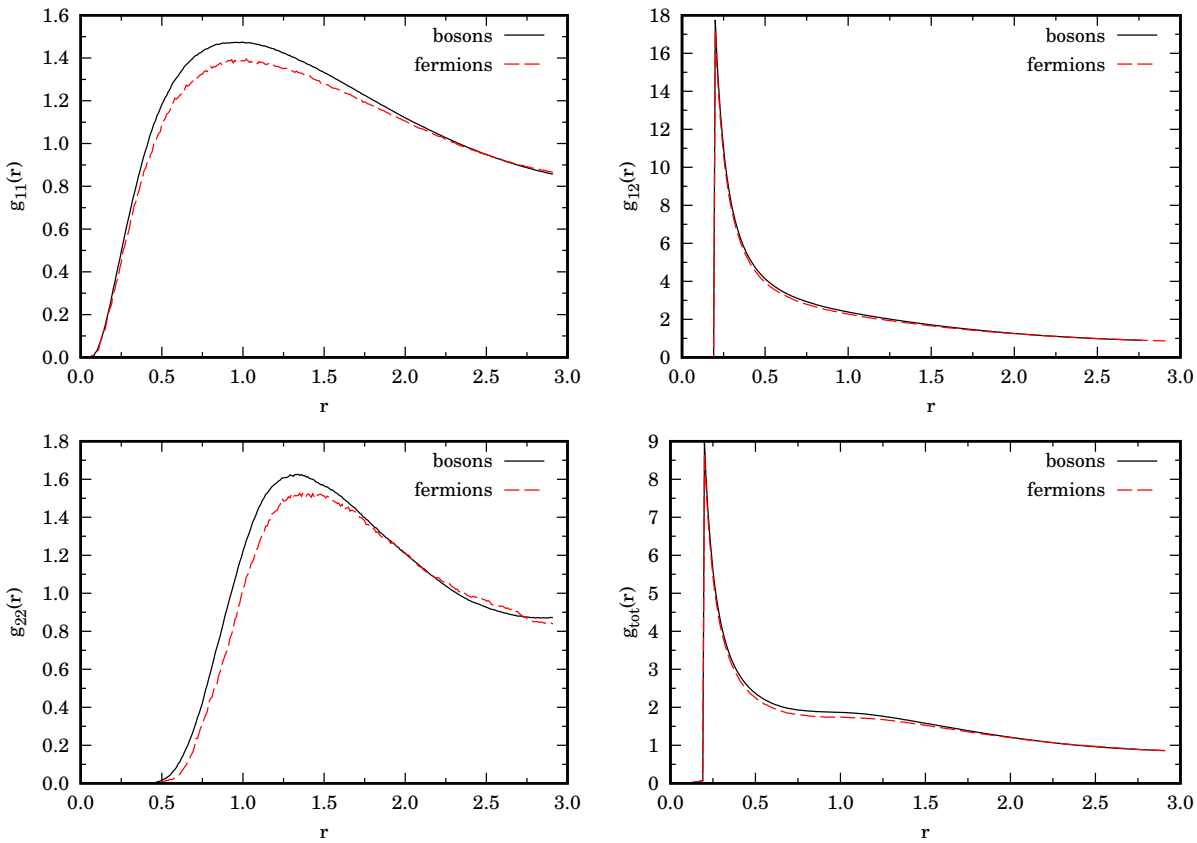

FIG. 1. The partial, g_{ij}, and total, g_{tot}, radial distribution functions for a binary mixture with $N = 20, x_1 = 1/2, \mu_1 = 1, \mu_2 = 1000, \sigma = 0.4$. The mixture is in a thermodynamic state with a reduced temperature $T = T'/\mathscr{T} = 0.1$ with T' measured in degrees Kelvin and a reduced number density $n = n'\mathscr{L}^3 = 0.1$ with n' measured in cm^{-3}. In the path integral we chose $M = 20$ and used either Bose-Einstein or Fermi-Dirac statistics among like particles. In the figures $r = r'/\mathscr{L}$ where r' is measured in centimeters.

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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None declared.

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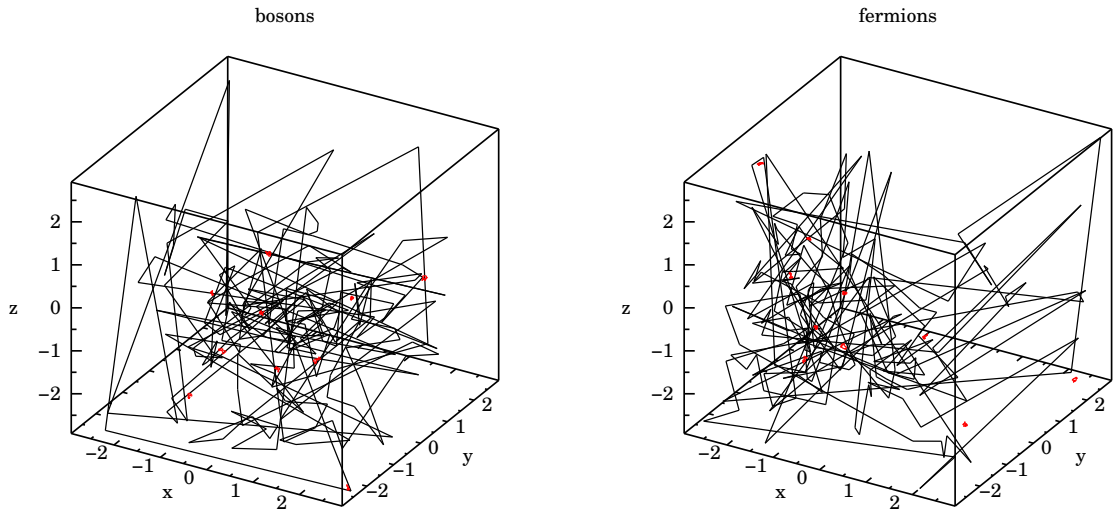

FIG. 2. Snapshots of the simulation of a binary mixture with $N = 20, x_1 = 1/2, \mu_1 = 1, \mu_2 = 1000, \sigma = 0.4$. The mixture is in a thermodynamic state with a reduced temperature $T = T'/\mathscr{T} = 0.1$ with T' measured in degrees Kelvin and a reduced number density $n = n'\mathscr{L}^3 = 0.1$ with n' measured in cm^{-3}. In the path integral we chose $M = 20$ and used either Bose-Einstein (left panel) or Fermi-Dirac (right panel) statistics among like particles. Red paths are for the heavy slow species and black paths are for the light fast species.

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